

W11 Pre-lab Quiz: IoT

Started: Mar 14 at 11:57am

Quiz Instructions



Week 11

Pre-lab Quiz



Instructions

Study all the materials for the module, then answer the questions.



Question 1 1 pts

What is a primary advantage of using Arduino boards in IoT education?



They are low-cost, programmable platforms ideal for prototyping



They offer high-performance cloud computing capabilities



They include built-in AI processors



They are designed for industrial-scale deployments



They require no programming knowledge



Question 2 1 pts

What is the primary goal of the OpenFog Reference Architecture?



To enable low-latency, reliable IoT applications through fog computing



To define cloud-based IoT services



To standardize IoT security protocols



To replace the IIRA model



To develop mobile IoT applications



Question 3 1 pts

Which of the following best describes the fourth generation of IoT systems?



Systems with semiautonomous smart objects and decentralized architectures



Systems based solely on RFID and WSN



Cloud-only systems with centralized control



Systems without edge computing



Systems limited to industrial automation



Question 4 1 pts

Which of the following is a key economic criterion for selecting a sensor?



Total cost of ownership (TCO)



Sensitivity to environmental noise



Signal-to-noise ratio



Response time



Operating temperature range



Question 5 1 pts

Which of the following best describes the relationship between IoT and Machine-to-Machine (M2M) communication?



IoT includes M2M but also supports other types of interactions



M2M is broader than IoT and includes cloud computing



IoT and M2M are completely unrelated



M2M only applies to industrial automation, not IoT



IoT is a subset of M2M focused on consumer devices



Question 6 1 pts

Which of the following is NOT a valid sensor selection criterion category?



Aesthetic design



Economic criteria



Environmental criteria



Sensor characteristics



Application requirements



Question 7 1 pts

What is the role of middleware platforms in IoT systems?



They connect devices to applications and manage data flow



They manufacture IoT hardware components



They provide physical interfaces for sensors



They replace the need for cloud computing



They act as user-facing dashboards



Question 8 1 pts

Which of the following is an example of a mechanical stimulus that a sensor might detect?



Pressure



Temperature



Magnetic field



Light intensity



Chemical concentration



Question 9 1 pts

Which of the following reference architectures decomposes IoT systems into five functional domains?



Industrial Internet Consortium Reference Architecture (IIRA)



OpenFog Reference Architecture



IoT-A Architectural Reference Model (ARM)



ISO/IEC CD 30141 IoT RA



Microsoft Azure IoT Reference Architecture



Question 10 1 pts

Which of the following best describes the Internet of Things (IoT) according to the ITU?



A global infrastructure enabling advanced services by interconnecting physical and virtual things



A system where the Internet is connected to the physical world via sensors



A network of smart devices using intelligent interfaces in smart spaces



A rebranding of ubiquitous computing



A cloud-based system for managing connected devices

Quiz saved at 12:04pm

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